Dead Time in Voltage Source Converters: Symmetric and Asymmetric Insertion, Harmonic Effects (with Emphasis on Even Harmonics) and Compensation Methods

Abstract

Dead time (blanking time) is the deliberate delay inserted between the turn-off of one switch of a phase leg and the turn-on of its complementary switch, and it is unavoidable in hard-switched voltage source converters (VSCs). Although it is a small fraction of the switching period, it introduces a systematic volt–second error that distorts the output voltage. This note reviews the origin of the error, the difference between symmetric and asymmetric dead-time insertion, and the resulting harmonic spectrum. Particular attention is given to the mechanisms that break the half-wave symmetry of the error voltage and therefore generate *even* harmonics and a DC component — effects that are often overlooked because the textbook analysis predicts odd harmonics only. Hardware and software compensation strategies are then surveyed at a system level, without entering into implementation details, and a comparison is given to support the selection of a suitable method. A comprehensive bibliography is provided.

Contents

1 Introduction **2**

2 Dead-time insertion strategies **2**
2.1 Nominal, effective and equivalent dead time 2
2.2 Symmetric versus asymmetric insertion 2

3 Average voltage error model **3**
3.1 Dead-time term 3
3.2 Conduction-drop term 4
3.3 Zero-current clamping and parasitic capacitances 4

4 Harmonic content: the symmetric case **4**

5 Mechanisms generating even harmonics and DC **5**
5.1 Displacement of the current zero crossings 5
5.2 Interaction between conduction drops and the modulating function 5
5.3 Upper/lower asymmetry combined with modulation 5
5.4 Asymmetric zero-current clamping and single-phase operation 6
5.5 Sequence components and system consequences 6

6 Hardware compensation and mitigation **6**

7 Software compensation **7**

8 Practical guidelines **9**

9 Conclusion 9

1 Introduction

In a hard-switched phase leg the two series-connected devices must never conduct simultaneously. Because turn-off is always slower than turn-on for minority-carrier devices, and because gate-driver propagation delays are not perfectly matched, a blanking interval T_d is inserted between the two gate commands. The interval must cover the worst-case turn-off storage and fall time, the spread of the driver propagation delays, and the tolerance of the modulator, over the full temperature and current range [1, 2, 3].

During this interval both devices are off and the leg output potential is not imposed by the modulator: it is determined by the sign of the load current, which freewheels through one of the antiparallel diodes. The resulting output voltage therefore differs from the commanded one by a pulse of amplitude equal to the DC-link voltage and duration equal to the dead time. Averaged over one modulation period, this becomes a current-polarity dependent voltage error that produces low-order harmonic distortion, torque ripple in motor drives, current distortion in grid-tied converters, and — at low modulation index — gain loss and even instability of the current loop [4, 6, 5, 28].

The classical treatment assumes that the error voltage is a square wave in phase with the load current and concludes that only odd harmonics (5th, 7th, 11th, 13th, ...) are produced. This conclusion holds only if the error waveform has half-wave symmetry. Several practical mechanisms destroy that symmetry, and the consequence is the appearance of even harmonics and of a DC component, which are far more troublesome than the odd ones because of transformer saturation, DC injection limits, and low-frequency torque pulsation. Section 5 is devoted to this point.

2 Dead-time insertion strategies

2.1 Nominal, effective and equivalent dead time

Let T_d be the nominal blanking time programmed in the modulator, t_{on} the turn-on delay (including driver propagation and current rise) and t_{off} the turn-off delay (including storage time and voltage rise). The volt–second error does not depend on T_d alone but on the *effective* dead time

$$T_d^{\text{eff}} \;=\; T_d + t_{on} - t_{off}, \tag{1}$$

which can be significantly larger or smaller than T_d and, for IGBTs, is strongly dependent on collector current and junction temperature [6, 9, 11, 26]. Because t_{on} and t_{off} of the upper and lower device are never identical — different gate resistors, different layout, different thermal state, device-to-device spread — the effective dead time of the positive-current commutation generally differs from that of the negative-current commutation:

$$T_d^{+} \neq T_d^{-}. \tag{2}$$

This is the first source of asymmetry, and it exists even when the modulator inserts a perfectly symmetric blanking time.

2.2 Symmetric versus asymmetric insertion

Two implementations are commonly found in modulators (Fig. 1):

- **Symmetric insertion.** The rising edge of each of the two complementary gate signals is delayed by $T_d/2$ (or the pulse of each device is shortened by $T_d/2$ at both edges). The

commanded pulse remains centred on the carrier reference; the modulation is unbiased in time but each pulse is shortened by T_d.

- **Asymmetric (one-edge) insertion.** The whole delay T_d is applied to the turn-on edge only, or is applied to one device only. The pulse centroid is displaced by $T_d/2$, which introduces an additional phase shift of the fundamental and, more importantly, an upper/lower asymmetry of the leg behaviour.

A further, often unintentional, source of asymmetry is the combination of the blanking logic with the sampling scheme: single-update (symmetric) versus double-update (asymmetric) regular sampling produces different residual errors, and asymmetric regular sampling introduces a sideband structure of its own [1, 16].

From the point of view of the harmonic content, what matters is not the label attached to the insertion scheme but whether the resulting volt–second error is the same, in magnitude and duration, for the two half-cycles of the load current.

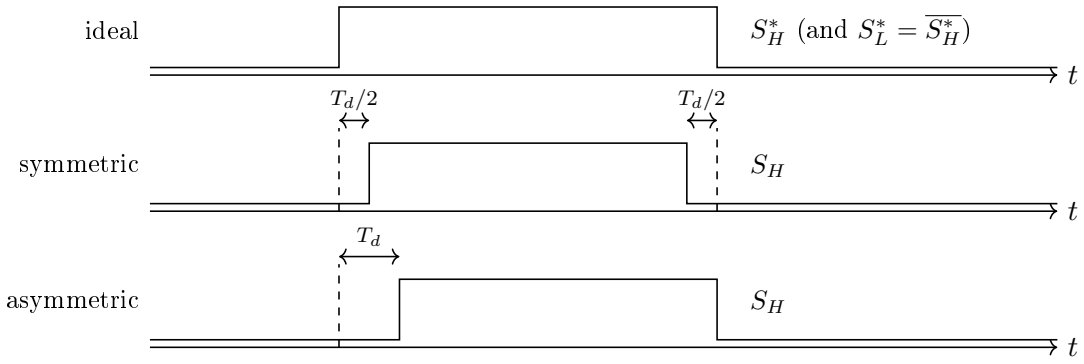


Figure 1: Symmetric (both edges shifted by $T_d/2$) versus asymmetric (turn-on edge only delayed by T_d) dead-time insertion. In both cases the on-pulse is shortened by T_d, but in the asymmetric case its centroid is displaced.

3 Average voltage error model

3.1 Dead-time term

Assume the load current i is approximately constant over one switching period T_s and that the leg output is clamped by the freewheeling diode during the blanking interval. The average error between the actual and the commanded leg voltage is

$$\Delta v \;=\; -\operatorname{sgn}(i)\,\frac{T_d^{\text{eff}}}{T_s}\,V_{dc} \;\triangleq\; -\operatorname{sgn}(i)\,V_\varepsilon, \tag{3}$$

i.e. the converter loses voltage when it sources current and gains voltage when it sinks it [6, 9, 11]. With $T_d = 3\,\mu\text{s}$, $f_s = 4\,\text{kHz}$ and $V_{dc} = 700\,\text{V}$ the error is about 8.4 V per leg, i.e. some 2 % of the fundamental at rated voltage but a much larger relative error at low speed or low modulation index.

If the two half-cycles see different effective dead times, (3) becomes

$$\Delta v \;=\; \begin{cases} -V_\varepsilon^{+}, & i > 0 \\ +V_\varepsilon^{-}, & i < 0. \end{cases} \tag{4}$$

3.2 Conduction-drop term

The on-state voltage drops of the transistor (V_{ce} or $R_{ds}i$) and of the diode (V_f) add a second error term. For $i > 0$, with duty ratio d of the upper device,

$$\Delta v_{cond} \;=\; -\Big[\, d\,V_{ce} + (1-d)\,V_f \Big] \;=\; -\left[\frac{V_{ce}+V_f}{2} + \frac{V_{ce}-V_f}{2}\,(2d-1)\right], \tag{5}$$

and the sign reverses for $i < 0$. The first term behaves like the dead-time term (a square wave in phase with the current); the second is proportional to the product of $\mathrm{sgn}(i)$ and the modulating function $(2d-1) \simeq m\sin\omega t$, and this product is the key to the even-harmonic generation discussed in Section 5. Note that in wide-bandgap converters the reverse-conduction drop during the dead time can be very large (3 V to 5 V for a GaN HEMT in third quadrant without a gate signal), which makes this term comparable with the dead-time term itself [29, 27].

3.3 Zero-current clamping and parasitic capacitances

Around the current zero crossing the ideal model fails. When $|i|$ is smaller than the ripple, the current changes sign within the switching period and the polarity-dependent error becomes ambiguous; furthermore, if the current is not sufficient to charge and discharge the device output capacitances within the blanking interval, the leg output voltage does not commutate at all and remains clamped at its previous value (zero-current-clamping, ZCC). The result is a flat region of several degrees around the zero crossing in which the voltage error is not a square wave but a smooth, current-dependent function [13, 25, 23]. The width of this region is not the same for the two crossings when the leg is thermally or electrically asymmetric, which is yet another source of even harmonics.

4 Harmonic content: the symmetric case

Consider a sinusoidal current $i = \hat{I}\sin(\omega t - \varphi)$ and a perfectly symmetric leg, $V_\varepsilon^+ = V_\varepsilon^- = V_\varepsilon$. The error voltage is a square wave of amplitude V_ε, synchronous with the current and phase-shifted by φ with respect to the voltage:

$$\Delta v(t) \;=\; -\frac{4V_\varepsilon}{\pi} \sum_{n=1,3,5,\ldots} \frac{1}{n}\,\sin\big[n(\omega t - \varphi)\big]. \tag{6}$$

The waveform has half-wave symmetry, $\Delta v(t + T/2) = -\Delta v(t)$, hence *only odd harmonics* are present, with amplitudes decaying as $1/n$. In a balanced three-phase three-wire system the triplen components are zero-sequence and do not drive current, so the dominant terms are the 5th, 7th, 11th, 13th, which appear as $6k$ torque ripple in a machine and as characteristic low-order distortion in a grid-tied converter [6, 9, 10, 1].

It is worth stressing an intermediate case. If the two half-cycles differ only in *amplitude* but not in *duration*, (4) can be rewritten as

$$\Delta v(t) \;=\; -\underbrace{\frac{V_\varepsilon^+ - V_\varepsilon^-}{2}}_{\text{DC}} \;-\; \underbrace{\frac{V_\varepsilon^+ + V_\varepsilon^-}{2}}_{\text{amplitude}} \cdot \mathrm{sq}(\omega t - \varphi), \tag{7}$$

i.e. an unbalanced amplitude produces a *DC offset plus odd harmonics*, but no even harmonics by itself. This is an important distinction: amplitude asymmetry alone is not sufficient to generate even orders; it is however sufficient to generate a DC voltage error, which then closes a feedback loop that does generate them.

5 Mechanisms generating even harmonics and DC

Even harmonics require the loss of half-wave symmetry in a stricter sense: the positive and negative intervals of the error waveform must have *different durations*, or the error must contain a term modulated by the fundamental itself. Four practical mechanisms are relevant.

5.1 Displacement of the current zero crossings

Let the positive interval of $\operatorname{sgn}(i)$ last $\pi+\delta$ instead of π, i.e. duty ratio $D=\frac{1}{2}+\frac{\delta}{2\pi}$. The Fourier coefficients of a ±1 wave with duty ratio D are

$$c_0 = 2D-1, \qquad |c_n| = \frac{4}{n\pi}\,\bigl|\sin(n\pi D)\bigr|, \tag{8}$$

so that for a small displacement δ

$$c_0 \simeq \frac{\delta}{\pi}\,V_\varepsilon, \qquad |c_{n,\text{even}}| \simeq \frac{2\delta}{\pi}\,V_\varepsilon \quad (n=2,4,6,\dots \text{ small}), \tag{9}$$

i.e. each low-order even harmonic is approximately *twice* the DC term and, unlike the odd ones, its amplitude does not decay as $1/n$ for the first few orders (Fig. 2). A zero-crossing displacement of only 7° already produces even harmonics of about 6 % of the fundamental error component.

The zero crossings are displaced whenever the load current contains a DC offset — from current-sensor offset or gain mismatch, from ADC offset, from an unbalanced modulator, or from a previous transient. This creates a *positive feedback loop*: the DC current offset displaces the crossings, the displacement produces a DC voltage error, and the DC voltage error drives further DC current through the small loop resistance. In transformer-coupled or transformerless grid-tied converters this mechanism is a well-documented cause of DC injection, transformer saturation and second-harmonic distortion [36, 37, 39, 38, 40].

5.2 Interaction between conduction drops and the modulating function

The second term of (5) is proportional to $\operatorname{sgn}(i)\cdot m\sin\omega t$. The product of two waveforms of odd harmonic content yields even orders and a DC component:

$$\operatorname{sq}(\omega t-\varphi)\cdot m\sin(\omega t) \;=\; \frac{2m}{\pi}\cos\varphi \;-\; \frac{2m}{\pi}\cos(2\omega t-\varphi)+\dots \tag{10}$$

Hence a transistor/diode drop mismatch ($V_{ce}-V_f$) generates a DC term and a second harmonic of comparable magnitude, $\approx \frac{m}{\pi}\,(V_{ce}-V_f)$, *even with a perfectly symmetric dead time and no current offset*. With $V_{ce}-V_f \approx 0.5\,\text{V}$ this is of the order of 0.16 V per leg: negligible at rated voltage, but comparable with the fundamental command at very low speed, and directly relevant for GaN-based converters where the third-quadrant drop is several volts [11, 26, 29].

5.3 Upper/lower asymmetry combined with modulation

When $T_d^+ \neq T_d^-$ (mismatched gate resistors, unequal driver delays, different junction temperatures of the upper and lower device, one-edge dead-time insertion), the DC term of (7) appears immediately. In a closed-loop drive the current controller reacts to it by biasing the modulating signal, which again multiplies a fundamental-frequency quantity by $\operatorname{sgn}(i)$ and produces even orders. In an open-loop V/f drive the DC voltage error directly produces a DC current, closing the same loop described above. Asymmetric leg cooling — very common in half-bridge modules with a hot upper device — makes this effect load- and time-dependent.

5.4 Asymmetric zero-current clamping and single-phase operation

The ZCC region described above is generally not symmetric about the two crossings, since the charge required to commutate the leg and the available current differ for the two transitions; the resulting error waveform has different positive and negative durations, producing even orders [13, 25]. In single-phase converters (and in three-phase converters with an unbalanced or four-wire load) there is no cancellation mechanism, so the DC and even components appear directly in the output current [36, 24].

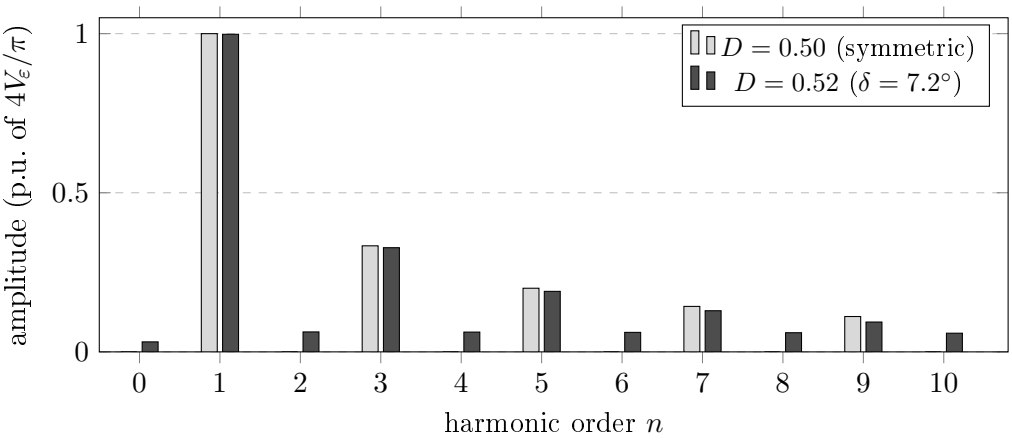


Figure 2: Spectrum of the dead-time error voltage for a symmetric error square wave and for an error wave whose zero crossings are displaced by 7.2°. The displacement introduces a DC component and even harmonics of nearly constant amplitude, at the expense of a barely visible change of the odd orders.

5.5 Sequence components and system consequences

In a balanced three-phase converter with identical legs, the harmonic sets of the error voltage have the following sequences: the DC component is zero-sequence (no current in a three-wire system, but a real DC magnetising bias if a transformer is present with a neutral path or in single-phase legs); the 2nd harmonic is *negative* sequence and therefore does drive current in a three-wire system; the 4th is positive sequence; the 6th is zero sequence. Consequences:

- **Machine drives.** The 2nd harmonic (negative sequence) maps into a 3ω torque ripple in the rotor frame, while a DC current component produces a 1ω ripple and a localised heating of one phase; the classical 5th/7th pair gives the well-known 6ω ripple [7, 12, 20].
- **Grid-tied converters.** DC injection is limited by grid codes and product standards (typically 0.5 % of rated current or less), and even harmonics are limited more severely than odd ones — IEEE Std 519 restricts them to 25 % of the adjacent odd limits [41, 43, 44, 42]. A DC bias saturates distribution transformers, increases their losses and noise, and further distorts the current [37].
- **DC-link and stability.** The distortion reflects into the DC-link current and interacts with the input filter, degrading damping and, at light load, causing low-frequency instability [28].

6 Hardware compensation and mitigation

The hardware approach either reduces the required dead time or removes its effect at the commutation itself.

- **Matched and optimised gate drives.** Reducing the spread of t_{on} and t_{off} between the two devices allows a shorter T_d with the same margin, and reduces the upper/lower asymmetry that generates DC and even orders. Negative gate bias, active Miller clamping and low-inductance layout suppress the cross-talk that would otherwise force a longer blanking time [27, 3].

- **Adaptive (self-adjusting) dead time.** The blanking interval is terminated as soon as the commutation is actually complete, detected through the collector–emitter (or drain–source) voltage, the dv/dt, or the desaturation comparator. The dead time then tracks the operating point instead of being set by the worst case [17, 18, 23].

- **Current-polarity-based gating (dead-time elimination).** If the sign of the leg current is known, only the device that must actually conduct is gated and the complementary one is kept permanently off; the freewheeling diode takes the current and no blanking interval is needed. A hysteresis band or a hybrid strategy is required around the zero crossing [17, 18, 22].

- **Soft-switching topologies.** In ZVS resonant-pole and auxiliary resonant commutated pole (ARCP) legs the output transition is imposed by the resonant network rather than by the device gating, and the dead-time distortion is largely removed at the price of an auxiliary circuit [31, 30].

- **Snubber and C_{oss} shaping.** Capacitive snubbers linearise the transition during the blanking interval; the voltage error becomes a smooth function of the current rather than a hard pulse, but the zero-crossing clamping region widens [25].

- **Fast wide-bandgap devices.** SiC and GaN allow dead times of tens of nanoseconds, so that T_d/T_s — and with it the whole distortion — is reduced by one or two orders of magnitude even at high switching frequency. Care is needed because the reverse-conduction drop during the dead time is high, making the residual error dependent on the dead time in a stronger, non-linear way [29, 27].

- **Analog pulse-correction circuits.** Historically, the missing volt–seconds were restored by hardware that lengthened the following pulse according to the detected current polarity [4, 5].

7 Software compensation

Software methods do not remove the error at the leg; they add a corrective term to the modulator reference so that the *average* output matches the command.

- **Average feed-forward compensation.** The simplest and most widespread scheme adds $+\mathrm{sgn}(i)\,V_\varepsilon$ to the voltage reference of each phase. It requires only the current sign and a value of V_ε; its weakness is precisely the sign detection near the zero crossings, where noise and ripple cause chattering [6, 8, 9, 7].

- **Improved polarity detection.** Hysteresis bands, use of the current *reference* instead of the measurement, use of an observed/estimated current, and smooth (piecewise-linear or arctangent) transition functions across the crossing reduce the residual distortion and, importantly, avoid introducing an artificial asymmetry between the two crossings [13, 15, 19].

- **Pulse-based compensation.** Instead of a mean value, each individual PWM pulse is corrected, which is more accurate at high switching frequency and short pulse widths and handles the minimum-pulse limitation explicitly [10, 16].

- **Look-up tables and self-commissioning.** The complete inverter non-linearity (dead time, conduction drops, clamping) is identified as a single current-dependent voltage curve, obtained by a standstill test or by online regression, and stored in a table. This captures the asymmetric part of the characteristic and is therefore effective against DC and even harmonics [21, 26, 14].

- **Disturbance observers and model-based estimation.** The distortion is treated as an unknown disturbance in the stationary or synchronous frame and estimated by an observer, then cancelled. No explicit dead-time model is needed [12, 20, 11].

- **Closed-loop harmonic regulators.** Proportional-resonant and vector-proportional-integral regulators tuned at the offending frequencies, multiple synchronous reference frames, repetitive control and iterative learning control suppress whatever periodic distortion is present, provided the loop bandwidth is sufficient. For even harmonics the resonant terms must be placed at 2ω and 4ω — in the synchronous frame at ω and 3ω — which is often forgotten when only the $6k$ terms are addressed [32, 33, 34, 35].

- **Explicit DC-component control.** A dedicated slow integral loop acting on the measured DC content of each phase current (or on an estimate of the transformer flux bias) breaks the positive feedback described in Section 5 and is the most direct remedy for DC injection and second-harmonic distortion [36, 37, 38].

- **Modulation-level remedies.** Mixed or hybrid PWM strategies, clamping of the leg carrying the largest current, and per-leg dead-time elimination based on the reference current sign reduce the number of commutations affected by the blanking time [22, 17].

- **Adaptive and learning schemes.** Recursive least squares, multiple linear regression on a physical model, and neural-network estimators adapt the compensation to temperature and ageing, which is relevant because T_d^{eff} drifts with the junction temperature [15, 26, 24].

A practical remark: any compensation scheme that applies the *same* correction to the two half-cycles cannot remove the DC and even-harmonic content produced by an asymmetric leg. Two separate compensation values, or a closed-loop DC regulator, are required.

Table 1: Qualitative comparison of the main compensation approaches.

Method	Principle	Extra HW	Needs $\operatorname{sgn}(i)$	Removes DC/even
Matched drivers, short T_d	reduce the cause	no	no	partly
Adaptive dead time	terminate blanking on dv/dt	yes	no	yes
Polarity-based gating	no blanking at all	yes	yes	partly
Soft switching (ARCP)	resonant transition	yes	no	yes
Average feed-forward	add $V_\varepsilon \operatorname{sgn}(i)$	no	yes	no
Pulse-based correction	per-pulse volt–second restore	no	yes	no
LUT / self-commissioning	identified non-linearity curve	no	yes	yes
Disturbance observer	estimate and cancel	no	no	yes
Resonant / repetitive control	closed-loop harmonic rejection	no	no	yes (if tuned)
Explicit DC loop	regulate DC current to zero	no	no	yes (DC only)

8 Practical guidelines

1. Size T_d on the worst case of turn-off delay and driver spread, but re-verify it at rated temperature: an unnecessarily conservative dead time is paid for at every switching event.
2. Characterise T_d^{eff} rather than T_d; measure it, if possible, as the offset of the voltage/current characteristic at standstill.
3. Verify separately the two half-cycles. A spectrum showing 2nd and 4th harmonics or a DC component is a symptom of leg asymmetry, sensor offset or asymmetric clamping — not of the dead time as such.
4. Check current sensor offset and gain matching before blaming the modulator: the two effects are indistinguishable in the spectrum, and they reinforce each other.
5. Use a weighted THD (WTHD) or the individual low-order limits of the applicable standard, not the plain THD, to judge the residual distortion [1, 41].
6. Combine a feed-forward term (fast, coarse) with a closed-loop harmonic or DC regulator (slow, accurate): the first handles the transient behaviour, the second removes the residual asymmetric content.

9 Conclusion

Dead time is a small quantity with a disproportionate effect on the low-frequency spectrum of a voltage source converter. The classical model, based on a symmetric error square wave synchronous with the load current, explains the 5th, 7th, 11th and 13th harmonics and the associated 6ω torque ripple. It does not explain the DC component and the even harmonics that are routinely observed on real converters. These arise when the half-wave symmetry of the error waveform is broken — by displaced current zero crossings (sensor offset or DC current), by the interaction between the transistor/diode drop mismatch and the modulating function, by unequal effective dead times of the upper and lower device, or by asymmetric zero-current clamping. Because even harmonics and DC are constrained much more tightly than odd harmonics, and because they feed back on themselves through the load, they deserve explicit attention. Hardware measures act on the cause (shorter and adaptive dead time, polarity-based gating, soft switching, wide-bandgap devices); software measures act on the effect (feed-forward, identified non-linearity tables, observers, resonant and repetitive control, dedicated DC loops). The two families are complementary, and only the methods that treat the two half-cycles independently are effective against the asymmetric part of the distortion.

References

[1] D. G. Holmes and T. A. Lipo, *Pulse Width Modulation for Power Converters: Principles and Practice*. Piscataway, NJ, USA: IEEE Press/Wiley, 2003.

[2] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications and Design*, 3rd ed. New York, NY, USA: Wiley, 2003.

[3] A. Wintrich, U. Nicolai, W. Tursky, and T. Reimann, *Application Manual Power Semiconductors*, 2nd ed. Ilmenau, Germany: SEMIKRON/ISLE, 2015.

[4] Y. Murai, T. Watanabe, and H. Iwasaki, "Waveform distortion and correction circuit for PWM inverters with switching lag-times," *IEEE Trans. Ind. Appl.*, vol. IA-23, no. 5, pp. 881–886, Sep. 1987.

[5] R. C. Dodson, P. D. Evans, H. T. Yazdi, and S. C. Harley, "Compensating for dead time degradation of PWM inverter waveforms," *IEE Proc. B*, vol. 137, no. 2, pp. 73–81, Mar. 1990.

[6] S.-G. Jeong and M.-H. Park, "The analysis and compensation of dead-time effects in PWM inverters," *IEEE Trans. Ind. Electron.*, vol. 38, no. 2, pp. 108–114, Apr. 1991.

[7] T. Sukegawa, K. Kamiyama, K. Mizuno, T. Matsui, and T. Okuyama, "Fully digital, vector-controlled PWM VSI-fed AC drives with an inverter dead-time compensation strategy," *IEEE Trans. Ind. Appl.*, vol. 27, no. 3, pp. 552–559, May/Jun. 1991.

[8] J.-W. Choi and S.-K. Sul, "A new compensation strategy reducing voltage/current distortion in PWM VSI systems operating with low output voltages," *IEEE Trans. Ind. Appl.*, vol. 31, no. 5, pp. 1001–1008, Sep./Oct. 1995.

[9] J.-W. Choi and S.-K. Sul, "Inverter output voltage synthesis using novel dead time compensation," *IEEE Trans. Power Electron.*, vol. 11, no. 2, pp. 221–227, Mar. 1996.

[10] D. Leggate and R. J. Kerkman, "Pulse-based dead-time compensator for PWM voltage inverters," *IEEE Trans. Ind. Electron.*, vol. 44, no. 2, pp. 191–197, Apr. 1997.

[11] A. R. Muñoz and T. A. Lipo, "On-line dead-time compensation technique for open-loop PWM-VSI drives," *IEEE Trans. Power Electron.*, vol. 14, no. 4, pp. 683–689, Jul. 1999.

[12] H.-S. Kim, H.-T. Moon, and M.-J. Youn, "On-line dead-time compensation method using disturbance observer," *IEEE Trans. Power Electron.*, vol. 18, no. 6, pp. 1336–1345, Nov. 2003.

[13] L. Ben-Brahim, "On the compensation of dead time and zero-current crossing for a PWM-inverter-controlled AC servo drive," *IEEE Trans. Ind. Electron.*, vol. 51, no. 5, pp. 1113–1118, Oct. 2004.

[14] A. Cichowski and J. Nieznanski, "Self-tuning dead-time compensation method for voltage-source inverters," *IEEE Power Electron. Lett.*, vol. 3, no. 2, pp. 72–75, Jun. 2005.

[15] N. Urasaki, T. Senjyu, K. Uezato, and T. Funabashi, "An adaptive dead-time compensation strategy for voltage source inverter fed motor drives," *IEEE Trans. Power Electron.*, vol. 20, no. 5, pp. 1150–1160, Sep. 2005.

[16] A. C. Oliveira, C. B. Jacobina, and A. M. N. Lima, "Improved dead-time compensation for sinusoidal PWM inverters operating at high switching frequencies," *IEEE Trans. Ind. Electron.*, vol. 54, no. 4, pp. 2295–2304, Aug. 2007.

[17] L. Chen and F. Z. Peng, "Dead-time elimination for voltage source inverters," *IEEE Trans. Power Electron.*, vol. 23, no. 2, pp. 574–580, Mar. 2008.

[18] Y.-K. Lin and Y.-S. Lai, "Dead-time elimination of PWM-controlled inverter/converter without separate power sources for current polarity detection circuit," *IEEE Trans. Ind. Electron.*, vol. 56, no. 6, pp. 2121–2127, Jun. 2009.

[19] S.-H. Hwang and J.-M. Kim, "Dead time compensation method for voltage-fed PWM inverter," *IEEE Trans. Energy Convers.*, vol. 25, no. 1, pp. 1–10, Mar. 2010.

[20] S.-Y. Kim, W. Lee, M.-S. Rho, and S.-Y. Park, "Effective dead-time compensation using a simple vectorial disturbance estimator in PMSM drives," *IEEE Trans. Ind. Electron.*, vol. 57, no. 5, pp. 1609–1614, May 2010.

[21] G. Pellegrino, P. Guglielmi, E. Armando, and R. I. Bojoi, "Self-commissioning algorithm for inverter non-linearity compensation in sensorless induction motor drives," *IEEE Trans. Ind. Appl.*, vol. 46, no. 4, pp. 1416–1424, Jul./Aug. 2010.

[22] Y. Wang, Q. Gao, and X. Cai, "Mixed PWM for dead-time elimination and compensation in a grid-tied inverter," *IEEE Trans. Ind. Electron.*, vol. 58, no. 10, pp. 4797–4803, Oct. 2011.

[23] Y. Park and S.-K. Sul, "A novel method utilizing trapezoidal voltage to compensate for inverter nonlinearity," *IEEE Trans. Power Electron.*, vol. 27, no. 12, pp. 4837–4846, Dec. 2012.

[24] M. A. Herrán, J. R. Fischer, S. A. González, M. G. Judewicz, and D. O. Carrica, "Adaptive dead-time compensation for grid-connected PWM inverters of single-stage PV systems," *IEEE Trans. Power Electron.*, vol. 28, no. 6, pp. 2816–2825, Jun. 2013.

[25] Z. Zhang and L. Xu, "Dead-time compensation of inverters considering snubber and parasitic capacitance," *IEEE Trans. Power Electron.*, vol. 29, no. 6, pp. 3179–3187, Jun. 2014.

[26] N. Bedetti, S. Calligaro, and R. Petrella, "Self-commissioning of inverter dead-time compensation by multiple linear regression based on a physical model," *IEEE Trans. Ind. Appl.*, vol. 51, no. 5, pp. 3954–3964, Sep./Oct. 2015.

[27] Z. Zhang, F. Wang, L. M. Tolbert, B. J. Blalock, and D. J. Costinett, "Active gate driver for fast switching and cross-talk suppression of SiC devices in a phase-leg configuration," in *Proc. IEEE Appl. Power Electron. Conf. (APEC)*, 2015, pp. 774–781.

[28] A. Guha and G. Narayanan, "Impact of dead time on inverter input current, DC-link dynamics, and light-load instability in rectifier-inverter-fed induction motor drives," *IEEE Trans. Ind. Appl.*, vol. 54, no. 2, pp. 1414–1424, Mar./Apr. 2018.

[29] A. Lidow, M. de Rooij, J. Strydom, D. Reusch, and J. Glaser, *GaN Transistors for Efficient Power Conversion*, 3rd ed. Chichester, U.K.: Wiley, 2019.

[30] W. McMurray, "Resonant snubbers with auxiliary switches," in *Conf. Rec. IEEE Ind. Appl. Soc. Annu. Meeting*, 1989, pp. 289–834.

[31] R. W. De Doncker and J. P. Lyons, "The auxiliary resonant commutated pole converter," in *Conf. Rec. IEEE Ind. Appl. Soc. Annu. Meeting*, 1990, pp. 1228–1235.

[32] R. Teodorescu, F. Blaabjerg, M. Liserre, and P. C. Loh, "Proportional-resonant controllers and filters for grid-connected voltage-source converters," *IEE Proc. Electr. Power Appl.*, vol. 153, no. 5, pp. 750–762, Sep. 2006.

[33] K. Zhou and D. Wang, "Digital repetitive controlled three-phase PWM rectifier," *IEEE Trans. Power Electron.*, vol. 18, no. 1, pp. 309–316, Jan. 2003.

[34] C. Lascu, L. Asiminoaei, I. Boldea, and F. Blaabjerg, "High performance current controller for selective harmonic compensation in active power filters," *IEEE Trans. Power Electron.*, vol. 22, no. 5, pp. 1826–1835, Sep. 2007.

[35] A. G. Yepes, F. D. Freijedo, Ó. López, and J. Doval-Gandoy, "Analysis and design of resonant current controllers for voltage-source converters by means of Nyquist diagrams and sensitivity function," *IEEE Trans. Ind. Electron.*, vol. 58, no. 11, pp. 5231–5250, Nov. 2011.

[36] G. Buticchi, E. Lorenzani, and G. Franceschini, "A DC offset current compensation strategy in transformerless grid-connected power converters," *IEEE Trans. Power Del.*, vol. 26, no. 4, pp. 2743–2751, Oct. 2011.

[37] G. Buticchi and E. Lorenzani, "Detection method of the DC bias in distribution power transformers," *IEEE Trans. Ind. Electron.*, vol. 60, no. 8, pp. 3539–3549, Aug. 2013.

[38] M. Armstrong, D. J. Atkinson, C. M. Johnson, and T. D. Abeyasekera, "Auto-calibrating DC link current sensing technique for transformerless, grid connected, H-bridge inverter systems," *IEEE Trans. Power Electron.*, vol. 21, no. 5, pp. 1385–1393, Sep. 2006.

[39] R. Sharma, "Removal of DC offset current from transformerless PV inverters connected to utility," in *Proc. 40th Int. Univ. Power Eng. Conf. (UPEC)*, 2005, pp. 1230–1234.

[40] F. Blaabjerg, R. Teodorescu, M. Liserre, and A. V. Timbus, "Overview of control and grid synchronization for distributed power generation systems," *IEEE Trans. Ind. Electron.*, vol. 53, no. 5, pp. 1398–1409, Oct. 2006.

[41] IEEE Recommended Practice and Requirements for Harmonic Control in Electric Power Systems, IEEE Std 519-2022.

[42] IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces, IEEE Std 1547-2018.

[43] Electromagnetic Compatibility (EMC) — Part 3-2: Limits — Limits for Harmonic Current Emissions, IEC 61000-3-2.

[44] Photovoltaic (PV) Systems — Characteristics of the Utility Interface, IEC 61727.